

Strong marine heatwaves trigger flowering in seagrass

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Abstract

In recent decades, the global intensification of marine heatwaves has impacted several ecosystems and species, including the endemic Mediterranean seagrass *Posidonia oceanica*. However, the scarcity of research in Eastern Mediterranean meadows, where historical and present thermal conditions differ from those of the Western Mediterranean, hampers our ability to draw comprehensive conclusions regarding the species' response to elevated sea temperatures. Here, we studied flowering patterns of *P. oceanica* meadows (3–15 m depth) of the Greek seas and assessed their potential association with marine heatwaves, while also examining the effects on plant growth associated with the transition from vegetative to sexual reproduction. To do so, we reconstructed flowering patterns from 2007 to 2021 across 58 meadows using lepidochronology and estimated summer marine heatwave metrics and categories using satellite sea surface temperature. The results revealed that flowering, while sporadic in most meadows (mean flowering frequency: 0.15), it was triggered by strong (Category II) summer marine heatwaves and a cumulative intensity exceeding 30°C days. Nevertheless, flowering density was low (mean flowering density: 11 inflorescences m⁻²) and not fuelled by frequency, intensity, or duration of marine heatwaves. Despite the potential long-term benefits of sexual reproduction, annual rhizome production of flowering shoots decreased 1.5-fold for at least 2 yr postflowering. This study uncovers the susceptibility of Eastern Mediterranean *P. oceanica* meadows, akin to their western counterparts, to marine heatwaves induced stress and its effects on reproductive strategies, raising concerns about the prospective plant growth and reproductive fitness to the escalating trend in marine heatwaves.

In the past few decades, the rise in global temperature has resulted in an increase in the occurrence of marine heatwaves (marine heatwaves) (Frölicher et al. 2018; Oliver et al. 2018)—defined as discrete, prolonged events of

anomalously high seawater temperature relative to a baseline period (climatology), occurring in a specific location and time of the year (Hobday et al. 2016). These events have significantly impacted the integrity of marine ecosystems globally (Oliver et al. 2019; Smale et al. 2019; Smith et al. 2023). The Mediterranean Sea, renowned for its ecological diversity and unique characteristics (Coll et al. 2010), is experiencing accelerated warming relative to other oceans globally, with a warming trend approximately three to four times higher than the global average (Pisano et al. 2020; Juza and Tintoré 2021). As a result, the Mediterranean Sea has become increasingly susceptible to more intense, long-lasting, and spatially extensive marine heatwaves (Garrahou et al. 2022; Pastor and Khodayar 2023). Future projections, even under the lowest carbon emission scenario (RCP 2.6), indicate that marine heatwaves will surpass the intensity and severity of present-day events (Darmaraki et al. 2019), exposing marine biota to unprecedented and potentially challenging conditions (Oliver et al. 2019; Smale et al. 2019).

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Additional Supporting Information may be found in the online version of this article.

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The impacts of warming have already become apparent across diverse taxonomic groups in the Mediterranean Sea (Coma et al. 2009; Garrabou et al. 2009, 2022), including the ecologically and economically important seagrass species *P. oceanica* (L.) Delile (Campagne et al. 2015), endemic to the Mediterranean Sea. Effects attributed to thermal stress in *P. oceanica* meadows include a decline in rhizome production (Mayot et al. 2005; Litsi-Mizan et al. 2023), enhanced leaf necrosis (Stipcich et al. 2022), a shift in nutritional value (Stipcich et al. 2023), and increased shoot mortality (Díaz-Almela et al. 2009; Marbà and Duarte 2010). In addition to disrupting physiological processes, thermal stress has also been shown to trigger the transition from vegetative to sexual reproduction, as evidenced by the extensive flowering reported after the intense marine heatwave events of 2003 in the Western Mediterranean (Díaz-Almela et al. 2007). The hypothesis of a thermal-induced reproductive response gained further support based on a mesocosm experiment where the flowering of *P. oceanica* shoots from the Western Mediterranean was triggered after exposure of the shoots to a simulated 6-week marine heatwave (Ruiz et al. 2018). Transcriptomic analysis provided molecular cues that the initiation of flowering could be attributed to a stress-induced response (Marín-Guirao et al. 2019). Similarly, transcriptomic analysis of inflorescences collected from the Western Mediterranean has identified genes implicated in flowering pathways responding to temperature cues (Entrambasaguas et al. 2017). Moreover, the recent extensive flowering observed across the Western Mediterranean (André et al. 2023; Stipcich et al. 2024; Tomas et al. 2024) following the intense and remarkably long marine heatwave of 2022 (Martinez et al. 2023) raises concerns about how *P. oceanica* reproductive dynamics will respond to the projected surge in marine heatwaves (Darmaraki et al. 2019).

Nevertheless, current knowledge primarily stems from Western meadows (refer to Supporting Information Fig. S1 and Tables S12 and S13 for an up-to-date spatial distribution of the research effort across the basin), which may not necessarily reflect the thermal response of the species across its entire distribution range. Divergent thermal regimes, evolutionary history, and local adaptations can yield distinct thermal responses within a species' geographic distribution (Bennett et al. 2019; Knies et al. 2009; Leaver et al. 2022; Nguyen et al. 2023). In the case of *P. oceanica*, meadows from the Western and Eastern basins, separated by a shallow ridge in the Strait of Sicily, exhibit contrasting thermal regimes. Historically, the Eastern basin has experienced higher temperatures compared with the Western counterpart (Pastor et al. 2020; Pisano et al. 2020), as well as an annual mean warming rate 1.3-fold faster between 1982 and 2020 (0.044 and 0.032°C yr⁻¹, respectively; Juza and Tintoré 2021). Indeed, warming rates during the summer period between 1982 and 2011 have been 2.6-fold faster in the Eastern basin compared with the Western basin (0.065 and 0.025°C yr⁻¹, respectively; Marbà et al. 2015). Moreover, the genetic

differentiation across the basin, likely shaped by the Pleistocene ice age and sustained by limited contemporary gene flow, suggests the possibility of vicariance divergence, leading to separate evolutionary trajectories for the meadows in each basin (Arnaud-Haond et al. 2007; Serra et al. 2010; Chefaoui et al. 2017). Therefore, it could be argued that Eastern Mediterranean meadows could possess heightened resilience owing to their natural history of prolonged exposure to elevated temperatures that favored the selection of more resilient genotypes and facilitated the adaptation to the local conditions. However, an opposing viewpoint could posit that the species' long generation time hampered rapid evolutionary adaptation (Aires et al. 2011; Arnaud-Haond et al. 2012), rendering the Eastern Mediterranean meadows more susceptible to climate warming as temperatures often exceed the species' thermal tolerance limits. To date, there is contrasting evidence about the thermal adaptation of the species. Studies in the Western Mediterranean suggested differential thermal responses between northern (colder) and southern (warmer) latitudes (Marín-Guirao et al. 2016). Moreover, morphological and genomic analyses showed differences in adaptation between meadows present inside a coastal lagoon characterized by extreme SSTs and salinity levels and those in the outer lagoon along the western coast of Sicily, Italy (Nguyen et al. 2023). However, recent transplantation studies, indicated that Western Mediterranean meadows perform similarly to Eastern Mediterranean ones under the thermal conditions of the latter (i.e., 5°C above the average SST in the Western Mediterranean), suggesting that thermal performance is not necessarily reflective of regional origin (Bennett et al. 2022). Additionally, evidence of a decline in annual rhizome production in Eastern Mediterranean meadows owing to rising temperatures (Litsi-Mizan et al. 2023) also challenges the possible higher thermal resilience or adaptation of Eastern *P. oceanica* meadows.

The phenomenon of flowering, triggered by thermal stress, represents a strategic shift in the reproductive strategy of seagrasses to effectively navigate evolving conditions and enhance their reproductive success (Marín-Guirao et al. 2019; Nguyen et al. 2021). Flowering enables the production of seeds that can be dispersed to new locations and thereby could constitute a strategy to promote the expansion and persistence of seagrass habitats by seeking out areas more suitable to their growth (Olesen et al. 2004; Short et al. 2007; Kendrick et al. 2012). Moreover, flowering introduces opportunities for genetic exchange, enhancing genetic diversity within a population (Procaccini et al. 2007), which holds particular importance when the species is facing thermal stress. If certain genotypes within a population demonstrate greater resilience to thermal stress, these advantageous traits can spread through natural selection, enhancing the population's overall adaptive capacity to global warming (Hughes and Stachowicz 2004; Waycott et al. 2006). Despite the aforementioned benefits, the transition from vegetative to sexual reproduction involves the reallocation of resources toward the

development and maintenance of reproductive structures, a process that can impact rhizome growth (Calvo et al. 2006). Consequently, while flowering constitutes a natural and adaptive process for the species, it occurs sporadically in *P. oceanica* meadows, whereas clonal expansion stands as the species' primary cost-effective propagation strategy (Hemminga and Duarte 2000). Hence, an imbalance between the two strategies, especially when confronted with heightened stress, might not prove advantageous for the species in the short term.

The study aimed to reconstruct long-term patterns of flowering in *P. oceanica* meadows within the understudied Greek seas and explore the hypothesized link between flowering events and marine heatwaves, while assessing the potential energetic costs associated with the transition from vegetative to sexual reproduction. To do so, we used lepidochronology, a standard reconstructive technique, to identify and date historical flowering events from 2007 to 2021 across 58 meadows. Concurrently, we used satellite sea surface temperature (SST) data during the same timeframe to detect the evolution of summer marine heatwaves across space and time and categorize the events based on their severity. By shedding light on the reproductive dynamics of *P. oceanica* meadows in an marine heatwave hotspot, this study contributes to a deeper understanding of the responses and adaptive strategies of this important marine species to changing environmental conditions.

Materials and methods

Study area and sampling procedure

A total of 58 monospecific *P. oceanica* meadows were sampled across Greece, encompassing the Ionian, Aegean, Cretan, and Levantine seas (Fig. 1) and covering a latitude range from 34° to 41° (refer to Supporting Information Fig. S1 for basin-scale spatial coverage). The region is characterized by annual mean SST (SST_{annual}) ranging from 17°C to 22.5°C, with higher values at lower latitudes (Litsi-Mizan et al. 2023). The meadows were distributed over both continental and insular areas, and the sampling water depths ranged from 3 to 15 m. Sampling surveys were conducted over a 11-yr period, from 2013 to 2023. Meadows from nearby areas were considered distinct sites when the distance between them was ≥ 1 km or when they were separated by significant geographical disruptions (e.g., enclosed bays). Scuba divers collected a total of ~ 3000 orthotropic shoots, with an average of 50 shoots per site ranging from 10 to 179 shoots (refer to Supporting Information Table S1 for further details). During collection, the orthotropic rhizome containing the living foliar shoot was carefully detached from the plagiotropic rhizome, allowing for accurate determination of the shoot age. Shoots were sampled at a distance of at least 1 m from each other to reduce the likelihood of gathering shoots connected to the same plagiotropic rhizome. Shoot density (number of shoots m^{-2}) was measured

three to five times at each site (with the exception of Afytos, Lesvos, Limnos, Strogili, and Trikeri) using a 40×40 cm² quadrat randomly placed over each meadow.

Laboratory data analysis

All collected *P. oceanica* shoots were carefully examined for the presence of floral stalk remnants along the rhizome. The year when the floral stalk remnant appeared at the rhizome, indicating the year of flowering, was estimated using the lepidochronology reconstructing technique described by Pergent (1990) and Pergent et al. (1989). This technique relies on identifying annual cyclic variations in the leaf sheath thickness along *P. oceanica* rhizomes. A lepidochronological cycle is characterized by two points of minimum sheath thickness, which typically occurs in late winter or early spring. Floral stalk remnants are commonly found amidst the thickest sheaths, usually observed during autumn (Pergent et al. 1989; Pergent 1990). The year of flowering and the age of the shoot were determined by counting the number of cycles in relation to the sampling year. To estimate the negative records of flowering (i.e., the number of non-flowering shoots), an average of 20 ± 3 shoots were also dated, while the remaining shoots were assumed to have the median age of the meadow. Additionally, during the lepidochronological analysis, the rhizome of all flowering shoots was cut into separate segments, each corresponding to a different lepidochronological year. The segments were dried at 60°C until a constant weight was reached. Annual rhizome production ($mg\ DW\ shoot^{-1}\ yr^{-1}$) was then estimated based on the dry weight of the rhizome segment per year. The rhizome segments corresponding to the most recent two growth cycles of each shoot were excluded, as the lepidochronological cycle could have been incomplete when shoots were collected (Peirano 2002; Litsi-Mizan et al. 2023). Data on annual rhizome production were retrieved from Litsi-Mizan et al. (2023) for some sites (Refer to Supporting Information Table S1 for further details).

Flowering metrics

Flowering metrics were estimated following Diaz-Almela et al. (2006, 2007), and are summarized in the Supporting Information Table S2. A minimum of 20 data points obtained from individual shoots were considered to retain a negative (absence) flowering record for a meadow in a given year. Conversely, the presence of a single flowering shoot was deemed sufficient to register a positive (presence) flowering record for that meadow in the respective year. Flowering intensity (FI, %) was estimated as the proportion of flowering shoots per meadow in a given year, and it was only calculated for years where a minimum of 50 data points per year were available. Flowering density (FD, inflorescences m^{-2}) was determined by multiplying the FI by the mean shoot density of the corresponding meadow, assuming that density remained constant throughout the years. In instances such as the Afytos and Trikeri sites, where shoot density data were absent, an

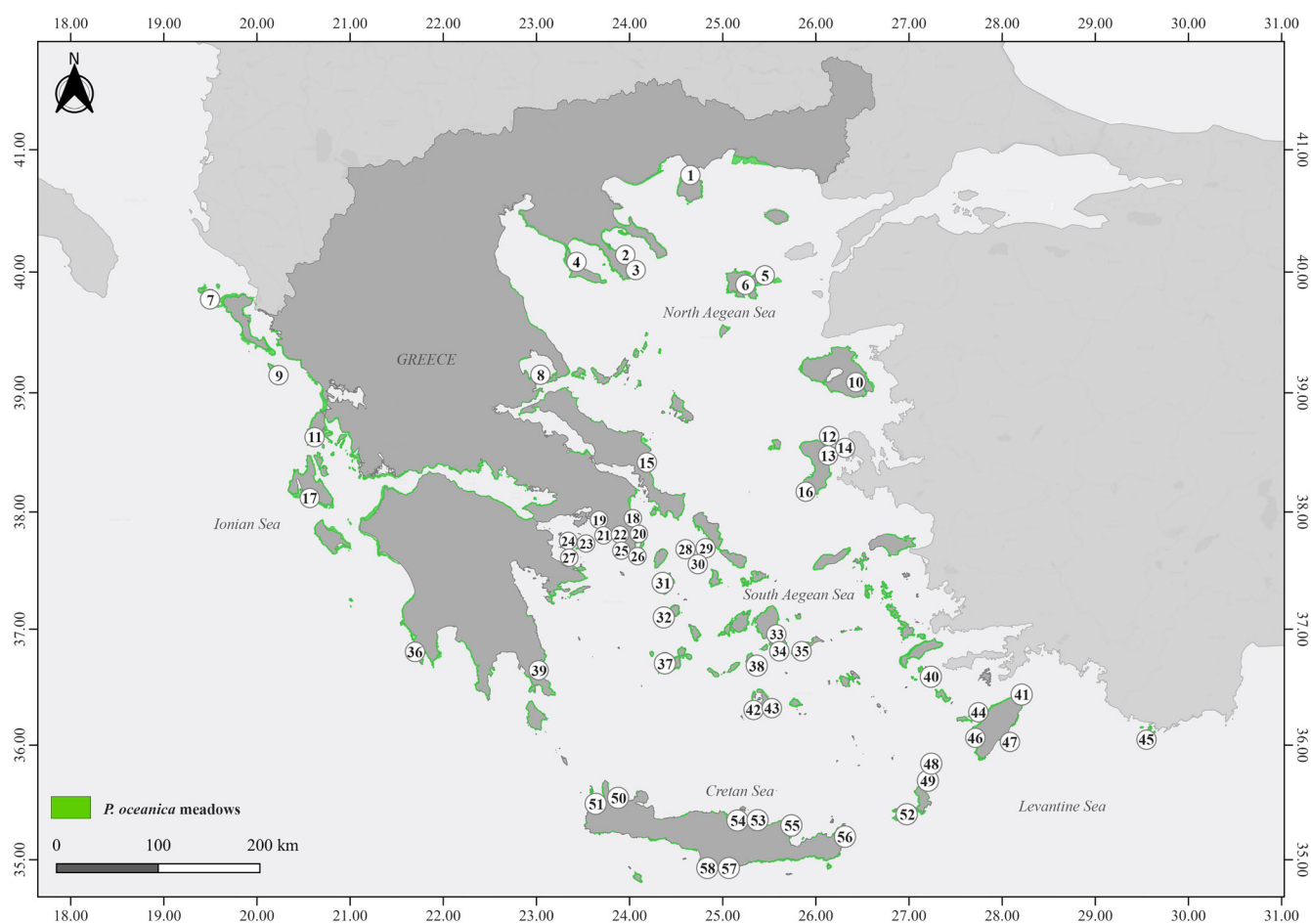


Fig. 1. Distribution of the sampled sites (circles) along the Greek Seas. The current *P. oceanica* distribution is indicated in green (from Panayotidis et al. 2022).

estimate was derived using the mean shoot density of the North Aegean region (333 number of shoots m^{-2} ; Gerakaris et al. 2021). Flowering frequency (FF) was estimated as the fraction of years that a meadow exhibited flowering, and it was only estimated for meadows with data points spanning at least four consecutive years. Flowering prevalence (FP, %) corresponded to the proportion of flowering meadows relative to the total number of meadows studied in any given year. As most of the reconstructed data fell within the years 2007–2021, all analyses were constrained to this temporal range (refer to the Results sections).

Thermal regime and summer marine heatwave metrics

To explore if marine heatwaves had an effect on flowering, surface seawater temperature (SST) satellite data from the Copernicus Marine Database (<https://marine.copernicus.eu/>) were used. The data consisted of daily, optimally interpolated (L4-REP), estimates of foundation SST (free or nearly free of any diurnal cycle, SST MED SST L4 Repobservations 010021) at 0.05° resolution grid covering the period from January 1, 1982 to December 31, 2021. For each site, the “nearest

neighbor” with available data was selected using MATLAB vR2023a. As the foundation SST estimates used in this study reflect the conditions within the top 10 m of the water column (Beggs 2021), we assumed that the satellite estimates accurately represented the environmental conditions at the specific sampling depth of each site that ranged from 3 to 15 m.

Daily foundation SST estimates served as the basis for characterizing the thermal regime across sites and throughout the years. This encompassed calculating the mean annual SST (SST_{annual}), maximum SST (SST_{max}), and summer SST (SST_{summer}). Additionally, the summer SST anomaly for each site and the mean across sites was calculated for the years 1982–2021, relative to the climatological average derived from the same period. Moreover, the identification of marine heatwaves events across sites and throughout the years followed the methodology outlined by Hobday et al. (2016) and was implemented in the R package *heatwaveR* (Schlegel and Smit 2018). The seasonal climatology and 90th percentile threshold were established based on a 39-yr period from 1982 to 2021 for each site, which was the longest possible period

(i.e., 1982 is the earliest SST data available for the region from Copernicus). The occurrence of a marine heatwave was recorded when the daily SST exceeded the 90th percentile for a minimum duration of five consecutive days, with a maximum gap of 2 d allowed to join separate events (Hobday et al. 2016). The analysis focused on the summer months (July, August, and September), corresponding to the warmest months of the year, and the period preceding the sexual reproductive phase of *P. oceanica* in early autumn. Accordingly, all relevant marine heatwave events falling within these summer months were aggregated to characterize the occurrence and magnitude of events within this specific timeframe for each year. The metrics encompassed: (1) the frequency (count) of events; (2) the maximum or peak intensity ($i_{\max}$), denoting the highest temperature anomaly value recorded during the summer events; (3) the mean intensity (i_{mean}), indicating the average temperature anomaly encompassing all marine heatwave days during the summer months; (4) the duration (days) of the events, representing the count of days surpassing the temperature threshold aggregated for the entire summer months; and (5) the cumulative intensity (i_{cum}), attained by integrating the intensity values over all summer marine heatwave days (Summarized in Supporting Information Table S3). Additionally, following Hobday et al. (2018), each summer event was classified based on the number of times the temperature anomaly exceeded the distance between the seasonally expected temperature and the 90th percentile threshold. A weak positive anomaly was considered a “moderate” (category I) event, whereas an anomaly twice as high as the aforementioned distance was considered a “strong” (category II) event, three times as “severe” (category III) event, and four times or greater as an “extreme” (category IV) event. If no marine heatwave event was detected at a specific site and year, it was treated as having a value of zero.

Statistical analysis

The statistical analyses utilized the reconstructed data of *P. oceanica* shoots, and the marine heatwave metrics and categories for the years 2007 to 2021. All computations were conducted using the R version 4.2.1 (R Core Team 2022), and all visual plots were generated using the *ggplot2* package (Wickham 2016).

Generalized Additive Mixed Models (GAMMs) were used to assess the effect of summer marine heatwaves and the geographic position (latitude), resembling differential annual thermal means, on *P. oceanica* flowering. The presence (1) or absence (0) of flowers at each site and year served as the response variable. Summer marine heatwave metrics (frequency, duration, i_{mean} , $i_{\max}$, and i_{cum}), and latitude, were included as explanatory variables. Given that marine heatwave metrics were correlated with each other (Pearson's coefficient > 0.6), separate models were fitted for each explanatory variables to avoid multicollinearity. The best model among significant relationships was determined based on the

lowest AIC (Akaike Information Criterion) (Akaike 1973) considering a difference of at least two. Additionally, the effect of marine heatwave events across different categories was evaluated using a GAMM model. The presence (1) or absence (0) of flowers at each site and year served as the response variable, while the explanatory variables included the frequency of marine heatwave events per category. For all GAMM models, the site was included as a random effect to account for repetitive measurements within each site and therefore potential autocorrelation. The binomial family with a complementary log-log (cloglog) link function was used, as it is suitable for modeling binary data with asymmetric distribution. GAMM models were fitted using the R packages *gam4* (Wood et al. 2017). Moreover, the effect of thermal stress on FD was assessed through a linear model, using marine heatwave metrics (frequency, duration, i_{mean} , $i_{\max}$, and i_{cum}) as the explanatory variables. Owing to the departure of the response variable from normal distribution (Shapiro–Wilk test, $p < 0.01$), a natural logarithm transformation was applied to achieve normality in the data. Given that marine heatwave metrics were correlated with each other (Pearson's coefficient > 0.6), separate models were employed for all explanatory variables to avoid multicollinearity. The models were fitted using the R package *MGCV* (Wood 2011).

To assess if the presence of flowers influenced the annual rhizome production of the shoot following a flowering event, a GLMM analysis was conducted. Rhizome segments were categorized into five periods relative to the flowering event: T0 was considered as the flowering year, T-2 and T-1 as the two consecutive years preceding the flowering event, and T1 and T2 as the two consecutive years following the event. Annual rhizome production (mg DW yr^{-1}) served as the response variable and periods were used as the explanatory variables with T-2 as the baseline. Additionally, the age of the shoot at each period was included as an explanatory variable to account for the potential effect of age on annual rhizome production (Tomasello et al. 2016). To account for the natural heterogeneity and to consider the temporal autocorrelation of the lepidochronological measurements on the same shoot, the shoot was included as a random effect. A gamma distribution with a logarithmic link was used as it suited the data after exploratory analysis. The model was fitted using the R package *MGCV* (Wood 2011).

Results

Flowering metrics

From the 2916 *P. oceanica* shoots analyzed, 144 shoots (4.9%) exhibited floral remnants (inflorescence). These flowering shoots were distributed across 42 out of the 58 surveyed meadows, representing 72.4% of the total sampled sites. The reconstructed dataset, derived from the lepidochronology analysis, spanned from 1993 to 2022 (Supporting Information Table S4), with the majority of data points (94%) concentrated

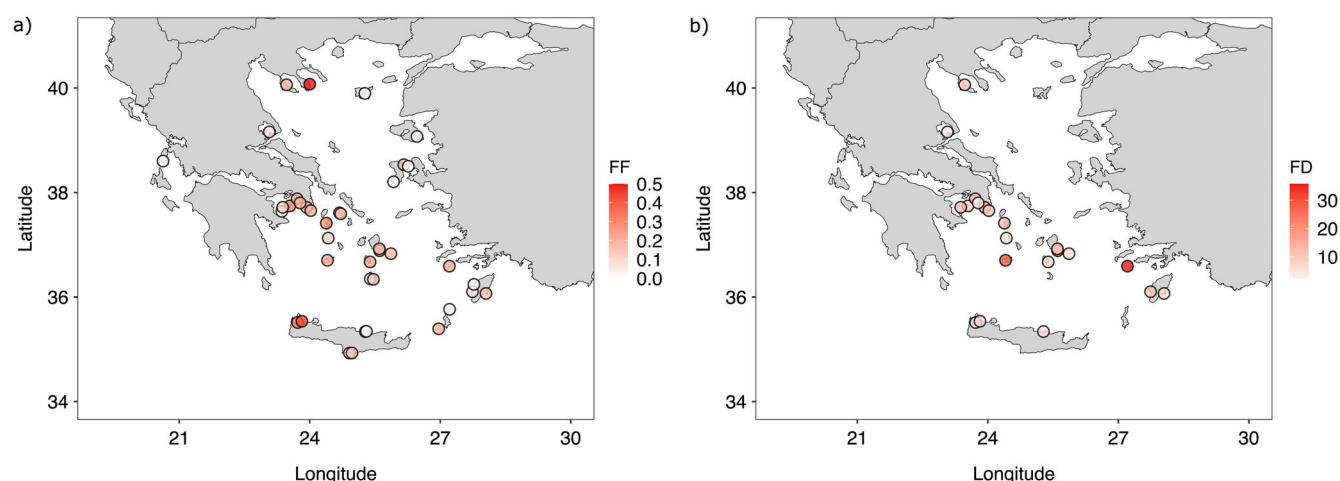


Fig. 2. (a) FF and (b) mean FD (inflorescence m^{-2}) of *P. oceanica* meadows in the Greek seas. Only sites with available data for these metrics are represented in the map. FF includes meadows with data points spanning a minimum of four consecutive years, while FD encompasses meadows with at least 50 data points for any given year.

between 2007 and 2021 (15 yr). To ensure a well-balanced representation for each year, the 2007–2021 timeframe was selected for all analytical assessments. During this period, 371 records (indicating presence or absence of flowering per meadow and year) were documented, of which 70 were positive (presence) and 301 negative (absence). The mean ($\pm$ SE) age of the shoots was 9.2 ± 0.3 yr and the age at which the

shoots flowered was 6.1 ± 0.4 yr, with a minimum age of 1 yr and a maximum age of 17 yr. The mean ($\pm$ SE) FF was 0.15 ± 0.02 , ranging from instances where flowering remained undetected for eight consecutive years to those where flowering occurred exceptionally often (FF = 0.43) (Fig. 2a). The mean ($\pm$ SE) FI was $2.5 \pm 0.3\%$, ranging from 0.6% to 8.9% and the mean FD was 11.2 ± 1.6 inflorescences m^{-2} ,

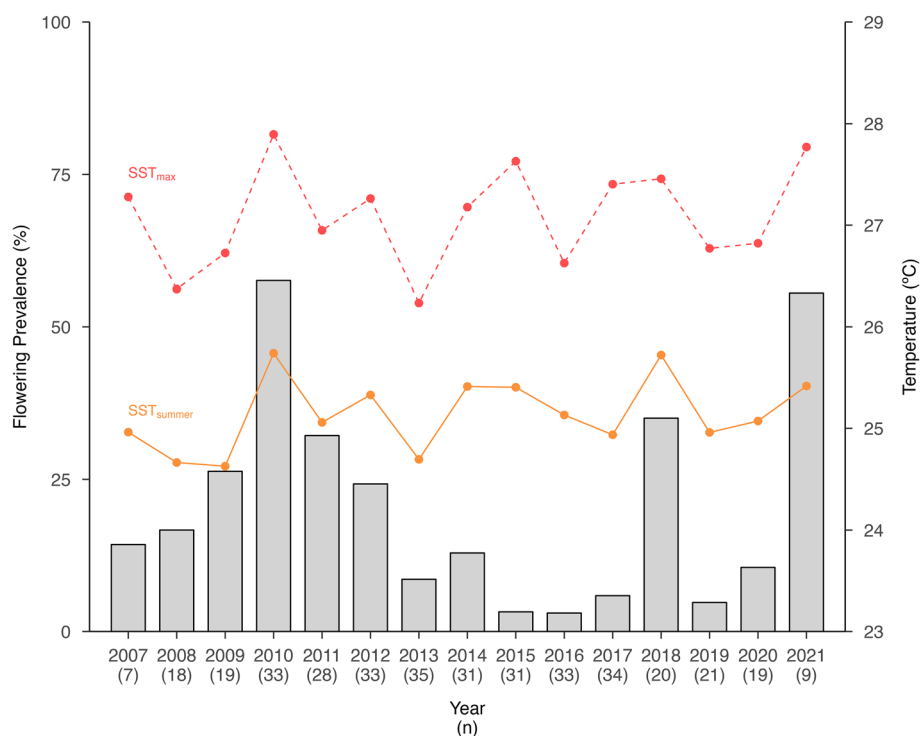


Fig. 3. FP (% of flowering meadows) of *P. oceanica* meadows from the Greek Seas for the years 2007–2021, together with the mean SST_{summer} and SST_{max} represented by the continuous (orange) and dashed (red) line, respectively. The total number of meadows (*n*) per year is represented in parenthesis below the corresponding year. For reference, FP in 2003 was 86% (Díaz-Almela et al. 2006).

ranging from 2.4 to 62.5 inflorescences m^{-2} (Fig. 2b). The annual mean ($\pm$ SE) FP was $20.7 \pm 4.6\%$, varying from 3.0% to 57.6%, with the highest FP occurring in 2010 followed by 55.5% in 2021 (Fig. 3). Refer to Supporting Information Tables S5–S7 for further details on flowering metrics results.

Thermal regime and summer marine heatwave metrics

Between 2007 and 2021, and across all sites, the global mean ($\pm$ SE) SST_{annual} of the study was $20.2 \pm 0.8^\circ C$, ranging from $17.9^\circ C$ to $22.9^\circ C$, while the global mean SST_{max} was $27.1 \pm 1.1^\circ C$, ranging from $23.8^\circ C$ to $30.4^\circ C$. In 2010, the highest temperatures were recorded with a mean SST_{annual} of $20.64^\circ C$ and a mean SST_{max} of $27.9^\circ C$. Conversely, the lowest temperatures were observed in 2007, with a mean SST_{annual} of $19.7^\circ C$, and in 2013, with a mean SST_{max} of $26.2^\circ C$. Across sites, the mean SST_{annual} exhibited a decrease toward higher latitudes, while no distinct trend was observed for the mean SST_{max} across different latitudinal positions (Supporting Information Fig. S2 and Tables S8–S10). Summer temperature anomalies consistently remained above average (positive) since 2007, reaching notable peaks in both 2010 and 2018 (Fig. 4). Regarding the aggregated summer marine heatwaves metrics, a mean ($\pm$ SE) frequency of events of 1.3 ± 0.2 per summer season was observed, with a mean i_{max} of $1.9 \pm 0.21^\circ C$, mean i_{mean} of $1.5 \pm 0.1^\circ C$, mean duration of 14.4 ± 3.2 d, and a mean i_{cum} of $29.2 \pm 6.5^\circ C$

days (Fig. 5). The year 2010 recorded the highest total summer i_{cum} with a mean of $77.0^\circ C$ days, closely followed by 2018 with $73.8^\circ C$ days. In 2017, the highest mean i_{max} of $3.5^\circ C$ was recorded; however, the duration was short, with a mean of 6.2 d during that summer. In 2021, there was the second-highest mean i_{max} of $3.2^\circ C$ and a mean duration of 18.3 summer days. Most events were categorized as moderate (Category I; 81.5%), followed by strong (Category II; 17.8%) and severe (Category III; 0.7%). No event was categorized as extreme (Category IV) for the studied sites and period. The mean ($\pm$ SE) frequency of moderate events per summer was 1.0 ± 0.03 , while for strong events was 0.3 ± 0.02 and for severe events, it was notably lower at 0.01 ± 0.003 events. The highest observed frequency for any given site or year was six for moderate events, four for strong events and one for severe events. Across the years, 2008, 2011, and 2013 were the only years that did not experience strong events, while the years 2010, 2017, and 2021 had the highest frequency of strong events, covering a spatial extent of at least 60% of the sites in each year. Moreover, severe events were observed exclusively in the years 2007 and 2017, constituting 1.7% and 6.9% of the sites, respectively (Fig. 5).

Effect of summer marine heatwaves on flowering

Based on the binomial GAMMs, concerning the aggregated summer marine heatwave metrics, duration, i_{max} , i_{mean} , and

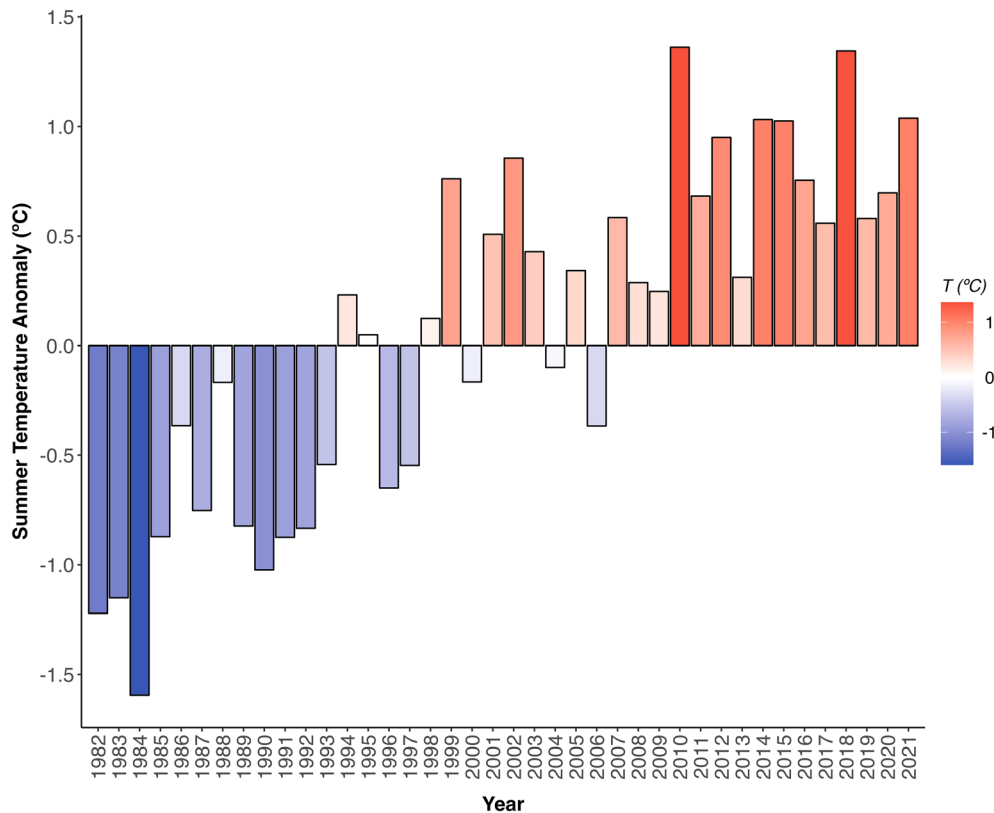


Fig. 4. Mean summer (July, August, and September) temperature anomalies across the studied sites for the years 1982–2021, relative to the climatological average derived from the same period. Blue colors represent negative anomalies, whereas red colors represent positive anomalies.

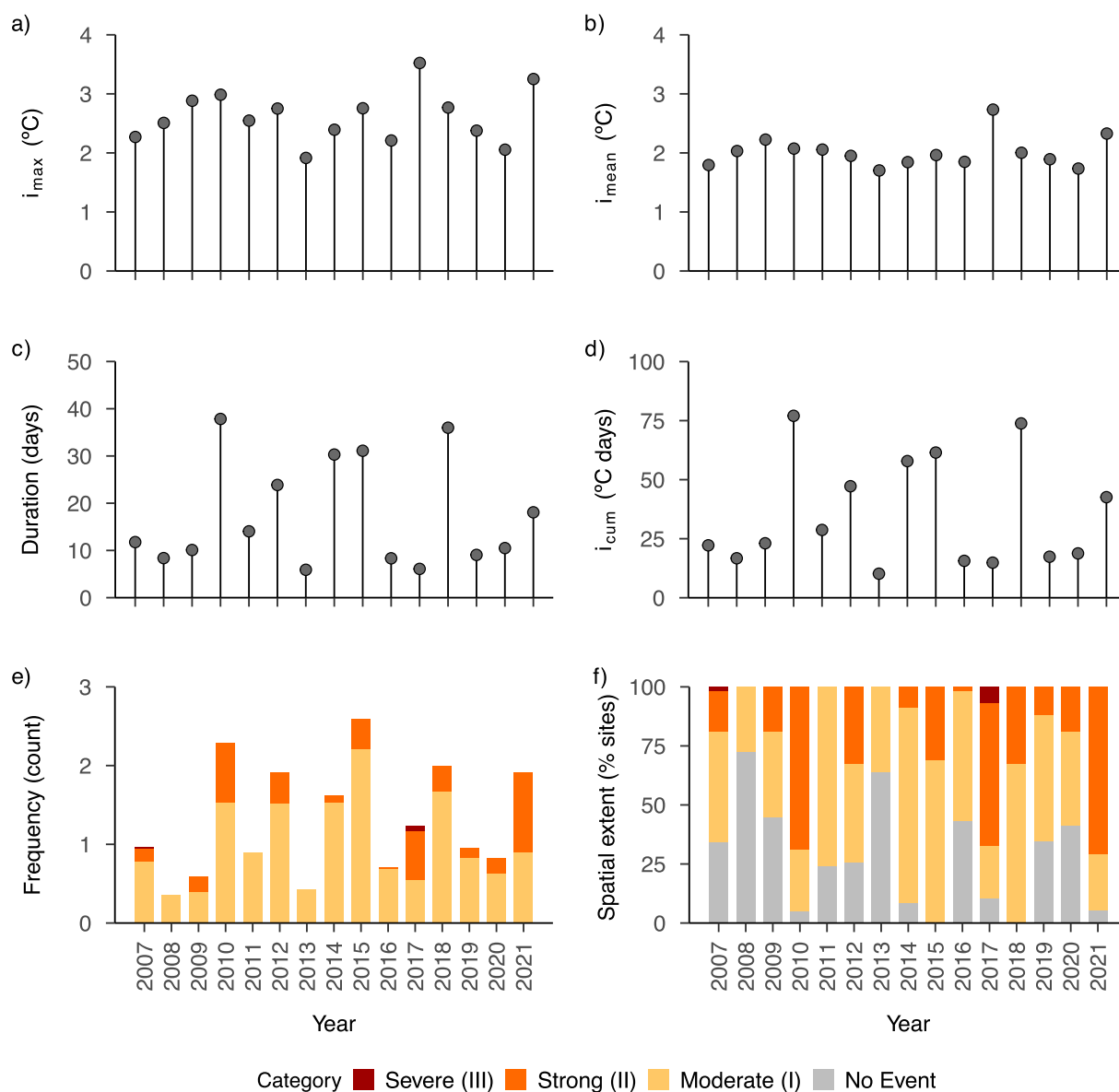


Fig. 5. Mean summer marine heatwave metrics across the studied sites for the years 2007–2021: (a) intensity max ($i_{\max}$ °C), (b) intensity mean (i_{mean} °C days), (c) duration (days), and (d) intensity cumulative (i_{cum} °C days). Summer marine heatwaves divided by categories: (e) mean frequency (count) per category and (f) spatial extent (% sites) based on the maximum category experienced during the summer season.

i_{cum} exhibited a significant effect on flowering occurrence (GAMM, $p < 0.05$; Supporting Information Table S11), whereas the frequency of summer marine heatwave events and geographic position of the meadow (latitude) did not yield a significant effect on flowering. Among all the significant metrics, i_{cum} was the most robust model based on the lowest AIC. Specifically, the probability of flowering increased over not flowering (positive) above an i_{cum} of 30 (°C days) (Fig. 6). Therefore, warmer conditions, as signified by higher i_{cum} (°C days), favor the occurrence of flowering within a given year in *P. oceanica* populations. Concerning the marine heatwave categories, the frequency of marine heatwaves categorized as

strong (Category II) exhibited a significant positive effect on flowering (GAMM, $p < 0.01$; Supporting Information Table S11), while the categories moderate (Category I) and severe (Category III) did not. Furthermore, FD remained unaffected by summer marine heatwaves (Linear Regression analysis, $p > 0.05$; Supporting Information Table S11).

Effect of flowering occurrence on annual rhizome production

The mean ($\pm$ SE) annual rhizome production for the flowering years (T0) was 35.5 ± 1.9 mg DW shoot⁻¹ yr⁻¹, whereas for the years preceding the flowering year (T-2 and

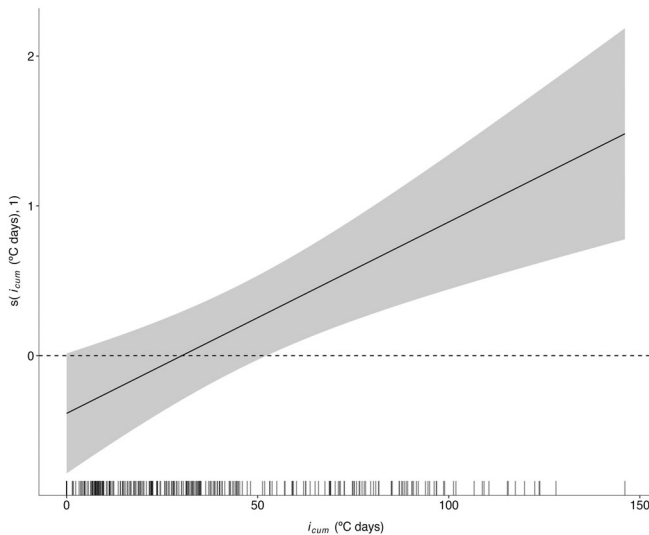


Fig. 6. Partial effect of total summer marine heatwaves cumulative intensity (i_{cum} °C days) on the probability of flowering of *P. oceanica* in any given year, as modeled by a GAMM. The x-axis denotes the range of total summer i_{cum} °C days, while the y-axis denotes the partial effect of i_{cum} °C days on the flowering probability, centered in zero. The dashed horizontal line at zero delineates a positive effect (above zero) or negative effect (below zero). The shaded areas indicate the upper and lower 95% confidence intervals. The lines at the bottom of the graph represent the number of samples at each predictor values.

T-1) it was 53.8 ± 3.7 and 48.2 ± 2.5 mg DW shoot⁻¹ yr⁻¹, respectively. For the 2 yr following the flowering event (T1 and T-2), the annual rhizome production was 36.4 ± 2.3 and 34.2 ± 3.3 mg DW shoot⁻¹ yr⁻¹, respectively (Fig. 7). The annual rhizome production 2 yrs preceding the flowering (T-2) was significantly higher than that of the flowering year (T0) and the subsequent 2 yr (T1 and T2) (GLMM, $p < 0.01$; Supporting Information Table S11), while no significant difference was observed between the 2 yr preceding the flowering event (T-2 and T-1). Additionally, the age of the shoot did not influence the vegetative growth in each period.

Discussion

In light of the ongoing surge in marine heatwave, this study revealed that while flowering remained sporadic across most meadows between 2007 and 2021, its incidence was particularly triggered by strong summer marine heatwaves, with i_{cum} exceeding 30°C days playing a fundamental role. Notably, a decline in annual rhizome production persisted for a minimum of 2 yrs following the flowering year, showcasing the energy costs of transitioning from vegetative to sexual reproduction. These findings presented in this study provide novel evidence of the susceptibility of Eastern Mediterranean meadows to marine heatwave-induced stress and its effects on reproductive strategies. Given the similarities in the thermal response of eastern and western counterparts, uncertainties

arise regarding the hypothesized variability in *P. oceanica* resilience to increasing temperatures across the basin.

The mean FF in the Eastern Mediterranean (0.15) aligned closely with that of the Western Mediterranean (0.17; Diaz-Almela et al. 2006) and with more recent findings from meadows along the Catalan coast (0.14; Jahnke et al. 2015). These patterns collectively underscore that sexual reproduction remains an occasional event in most *P. oceanica* meadows. In parallel, the mean FP in the Eastern Mediterranean between 2007 and 2021 (20.7%) closely resembled that of the Western Mediterranean between 1979 and 2004 (17%; Diaz-Almela et al. 2006). However, no year matched the remarkable FP of 86% recorded in 2003 (Diaz-Almela et al. 2006), except for 2010, which came closest to this exceptional level with a prevalence of 57.6%. Differences in flowering patterns across the basin were primarily observed in the form of reduced flower abundance, evident from both lower overall averages and lower upper limits, with no meadows exceeding an FD of 62 inflorescences m⁻² or an FI of 10%. Specifically, the mean FI across the Eastern Mediterranean ($2.5 \pm 0.3\%$) was 3.2-fold lower than the mean FI previously estimated for the entire Mediterranean Basin ($8 \pm 1\%$; Diaz-Almela et al. 2007), and 4.4-fold lower than previously reported for the Western Mediterranean ($11 \pm 2\%$; Diaz-Almela et al. 2006). These differences in flower abundance can be partially attributed to the exceptional flowering event of 2003 in the Western Mediterranean, which was unprecedented in its spatial extent and intensity. This event registered a mean FI of $22 \pm 4\%$ for the Western Mediterranean, including areas with remarkably high values such as the Balearic Islands, averaging $37 \pm 12\%$ (Diaz-Almela et al. 2006). Consequently, when excluding the 2003 event, the mean FI for the Western Mediterranean meadows decreases to 5%, drawing closer to the FI observed in the Eastern Mediterranean. Apart from the differences among the subregions of the basin, the considerable differences among individual meadows underline the significant spatial and temporal variations that characterize the overall flowering abundance of *P. oceanica* meadows.

While flowering occurred irregularly in the majority of meadows during the study period, its prevalence was notably shaped by marine heatwaves. Despite the significant influence of i_{max} , i_{mean} , and duration on the occurrence of flowering, the integrated cumulative thermal stress—measured by the i_{cum} —was the most robust model and highlights its pivotal role in predicting the flowering probability of *P. oceanica* meadows. Initially, the results observed for the marine heatwave categories seemed contradictory, as strong (Category II) events were positively associated with flowering, whereas severe (Category III), despite being warmer, did not exhibit a positive association with flowering. This discrepancy could be attributed to the infrequent occurrence of events that fell into the severe category between 2007 and 2021, constituting only 0.7% of events, coupled with their limited duration, averaging 7 d. The relevance of cumulative thermal stress is consistent

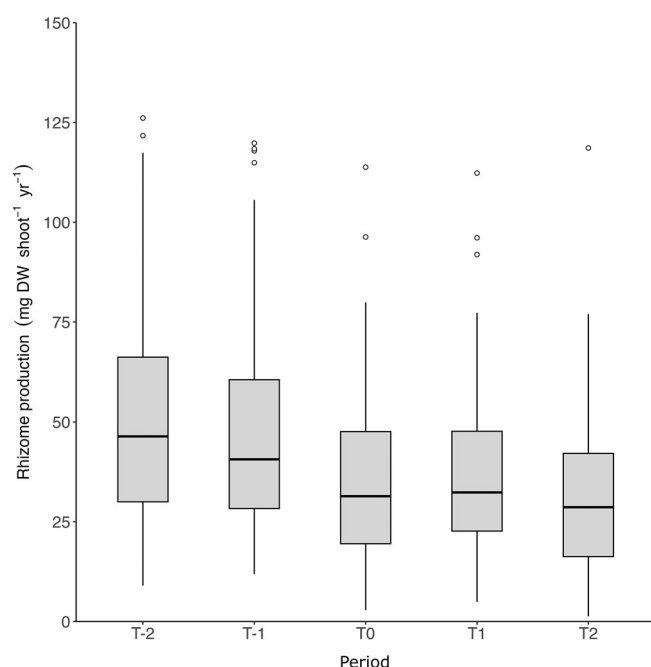


Fig. 7. *P. oceanica* annual rhizome production (mg DW shoot⁻¹ yr⁻¹) of flowering shoots across time (T-2 and T-1: two consecutive years before flowering, T0: flowering year, T1 and T2: two consecutive years after flowering). The boxes encompass the 25% and 75% quantiles, and the black lines represent the medians. The circles show data points beyond the whiskers.

with flowering patterns observed in the Western Mediterranean. Notable flowering events in this region, such as those in 2003 and 2022 (Díaz-Almela et al. 2007; André et al. 2023), coincided with severe summer marine heatwaves, with corresponding mean i_{cum} values of 56.5°C and 38.8°C days (Martínez et al. 2023). Similarly, the integrated cumulative thermal stress recorded during 2010 and 2021 in the Eastern Mediterranean also corresponded to noteworthy flowering records, which were associated with summer seasons marked by strong events, with aggregated mean i_{cum} values of 77.8°C and 36.4°C days, respectively. Moreover, the significance of cumulative heat stress exposure was previously emphasized by a mesocosm study that revealed activation of stress-related and flowering-inducing genes after 2 weeks of heat exposure, resulting in visible inflorescences after six weeks (Marín-Guirao et al. 2019). Conversely, a study with shorter (5 d) but similarly intense heat exposure did not yield evident signals of flower induction, with plants reverting to their normal state once the stress subsided (Marín-Guirao et al. 2016), highlighting the synergic importance of duration and intensity of heat stress on plant response.

At the basin scale, the similar stress-induced reproductive response of Eastern and Western Mediterranean meadows to marine heatwave suggests that despite the different thermal regimes (Juza and Tintoré 2021) and evolutionary histories across the basin (Arnaud-Haond et al. 2007; Serra et al. 2010;

Chefaoui et al. 2017), the Eastern Mediterranean ones are also vulnerable to the impacts of marine heatwaves, thus supporting previous findings on the vulnerability of these meadows to increasing temperatures (Litsi-Mizan et al. 2023) and marine heatwaves (Stipcich et al. 2022). This is further supported by the lack of significant influence of latitude on the flowering occurrence and the synchrony of flowering across a large geographic area, including northern (colder) and southern (warmer) latitudes and among Ionian, Aegean, Levantine, and Cretan seas. While we do not rule out the possible evolution of adaptive mechanisms for other biological processes, it is worth noting that the exceptionally long generation time and sluggish evolutionary rates of *P. oceanica* can considerably impede the pace of evolutionary adaptations (Aires et al. 2011; Arnaud-Haond et al. 2012). In the Greek seas, the higher clonality and diminished genetic diversity observed in northern Greek populations suggest a more recent establishment compared with their southern counterparts, likely a consequence of historical events such as the Last Glacial Maximum (Litsi-Mizan et al. 2024). Therefore, northern populations, with lower SST_{mean} compared to southern populations, may still retain signatures of their past environmental context. Indeed, although *P. oceanica* and its congeneric counterpart, *P. australis*, diverged 68 Myr before present (Aires et al. 2011), their timing of flowering coincides in time, albeit occurring in opposite seasons (Autumn and Spring, respectively; Hemminga and Duarte 2000), suggesting a highly conservative reproductive strategy in the genus *Posidonia*.

Apart from the crucial role that thermal stress plays in triggering flowering of *P. oceanica* across the basin (Díaz-Almela et al. 2007; Jahnke et al. 2015; Ruiz et al. 2018; Marín-Guirao et al. 2019), additional factors may mediate flowering occurrence and explain the occasionally low FP (e.g., in 2015) when i_{cum} was high. Shoot age (Balestri and Vallerini 2003), level of fish herbivory (Planes et al. 2011), resource availability, and genetic components like heterozygosity, relatedness and genotypic richness (clonality) (Jahnke et al. 2015) have been recognized as drivers influencing flowering patterns. Recent studies on Eastern Mediterranean meadows reveal higher genotypic richness (lower clonality) (0.8 on average for Greece; Litsi-Mizan et al. 2024, and 0.9 for Turkey and Cyprus; Tutar et al. 2022) compared to Western Mediterranean counterparts (0.6 on average with values as low as 0.1; Serra et al. 2010). Considering the negative association between genotypic richness and flower abundance as observed by Jahnke et al. (2015), it seems plausible that the higher genotypic richness observed in the Eastern Mediterranean meadows may play a role in their relatively lower levels of flower abundance (inflorescence m⁻²) compared with the Western meadows. Conversely, an alternative scenario is also possible, where higher genotypic richness might stem from historically frequent and/or high flower abundance events, provided these sexual reproduction events were successful. A direct assessment of clonality within flowering meadows would be necessary to establish this

connection, as well as the potential synergetic effect of the other aforementioned factors. On the other hand, despite the acquisition of long-term flowering data for the Eastern Mediterranean, it is important to acknowledge that the inherent limitations of lepidochronology might introduce certain ambiguities. This method enabled the reconstruction of an average of 8 yr per meadow, thereby introducing some imbalance in the data coverage per meadow throughout the entire 15-yr study period and potentially underestimating FP for certain years.

While looking at the energetic implications of flowering, the lower annual rhizome production observed during the flowering year and the subsequent 2 yr highlights the significant energetic cost associated with flowering in *P. oceanica*. Although previous studies showed a decrease in vegetative growth in shoots older than 2–4 yr (Tomasello et al. 2016), the lack of correlation between shoot age (for the year preceding and following flowering) and vegetative growth across the meadows surveyed suggests that the effect of flowering was more influential than aging. These findings align with the trade-off theory observed in many plant species, where resources are redirected from growth to support the energetic demands of flower, pollen, and seed production and maintenance (Van Drunen and Dorken 2012). In some plants, the reproductive organs can photosynthesize, acting as additional carbon resources and thereby, contributing toward the costs of their construction and maintenance (Bazzaz et al. 1979; Aschan and Pfanz 2003). In the case of *P. oceanica*, the activation of genes related to photosynthetic processes in female flowers suggests they are photosynthetically active (Entrambasaguas et al. 2017). However, as evidenced by the lower annual rhizome production, the resource acquisition of female flowers is not sufficient to sustain the metabolic costs of both female and male flowers, which may be also related to the low light availability during autumn and winter (Alcoverro et al. 1998), when flowers are produced. Our results align with studies in the Western Mediterranean, where a decrease in annual rhizome production during the years following the reproductive event was observed (Calvo et al. 2006; Montefalcone et al. 2013; Jahnke et al. 2015). However, they differ from Balestri and Vallerini (2003), who found no impact of flowering on short-term growth (rhizome elongation and number of leaves) but were based on observations from a single site. Owing to the age of the shoots at the time of flowering and the exclusion of the last two lepidochronological years due to the limitations of the method, we had limited data on annual rhizome production beyond T2 (i.e., more than 2 yr following the flowering event), which hindered our ability to assess the recovery of vegetative growth. Examining shoot recovery would provide valuable insights into the total energetic impact associated with shifting to sexual reproduction and its effects on the lifespan of the flowering shoot.

Given the predicted rise in marine heatwaves—particularly those of higher categories and prolonged durations

(Darmaraki et al. 2019)—a future increase in FF within *P. oceanica* meadows cannot be discounted. While such an uptick could potentially benefit the long-term species resilience by promoting genetic diversity, adaptation, and new meadow establishment (Olesen et al. 2004; Procaccini et al. 2007; Short et al. 2007; Kendrick et al. 2012), its potential drawbacks include the resource demands for sexual reproduction. This consideration gains significance in light of the anticipated declines in vegetative growth owing to warming (Stipich et al. 2022; Litsi-Mizan et al. 2023) and other anthropogenic pressures (Ruiz and Romero 2003; Marbà et al. 2006; McDonald et al. 2023). Increasing the monitoring effort on flowering and the less-explored subsequent stages (including pollination, fruit development, and seedling establishment) through in situ methods and lepidochronology can furnish more empirical insights into the reproductive dynamics of the species. Additionally, attention should also be directed towards identifying ecological tipping points and exploring the less understood dimensions of post-marine heatwave recovery trajectories. This, in turn, can enhance our ability to predictive and manage the diverse impacts posed by climate change on *P. oceanica* meadows, thereby facilitating more effective conservation strategies.

Data availability statement

The seagrass data that supports the findings of this study are available in the Supporting Information of this article and in the public repository EMODNET.

References

- Aires, T., N. Marbà, R. Cunha, G. Kendrick, D. Walker, E. Serrão, C. Duarte, and S. Arnaud-Haond. 2011. Evolutionary history of the seagrass genus *Posidonia*. *Mar. Ecol. Prog. Ser.* **421**: 117–130. doi:10.3354/meps08879
- Akaike, H. 1973. Information theory and an extension of the maximum likelihood principle, p. 267–281. *In* Proceedings of the 2nd International symposium of information theory. Akademiai Kiado.
- Alcoverro, T., M. Manzanera, and J. Romero. 1998. Seasonal and age-dependent variability of *Posidonia oceanica* (L.) Delile photosynthetic parameters. *J. Exp. Mar. Biol. Ecol.* **230**: 1–13. doi:10.1016/S0022-0981(98)00022-7
- André, S., and others. 2023. The 2022 mass flowering of *Posidonia oceanica* in the French Mediterranean Sea: Is it unprecedented? *Scientific Reports of Port-Cros National Park*, p. 37.
- Arnaud-Haond, S., and others. 2007. Vicariance patterns in the Mediterranean Sea: East–west cleavage and low dispersal in the endemic seagrass *Posidonia oceanica*. *J. Biogeogr.* **34**: 963–976. doi:10.1111/j.1365-2699.2006.01671.x

- Arnaud-Haond, S., C. M. Duarte, E. Diaz-Almela, N. Marbà, T. Sintes, and E. A. Serrão. 2012. Implications of extreme life span in clonal organisms: Millenary clones in meadows of the threatened seagrass *Posidonia oceanica*. *PLoS ONE* **7**: e30454. doi:10.1371/journal.pone.0030454
- Aschan, G., and H. Pfanz. 2003. Non-foliar photosynthesis—A strategy of additional carbon acquisition. *Flora: Morphol. Distrib. Funct. Ecol. Plants* **198**: 8–97. doi:10.1078/0367-2530-00080
- Balestri, E., and F. Vallerini. 2003. Interannual variability in flowering of *Posidonia oceanica* in the North-Western Mediterranean Sea, and relationships among shoot age and flowering. *Bot. Mar.* **46**: 525–530. doi:10.1515/BOT.2003.054
- Bazzaz, F. A., R. W. Carlson, and J. L. Harper. 1979. Contribution to reproductive effort by photosynthesis of flowers and fruits. *Nature* **279**: 554–555. doi:10.1038/279554a0
- Beggs, H. 2021. Temperature. Ch 14 in earth observation: Data, processing and applications. 3B—Surface Waters. CRCSE, Melbourne. pp. 245–279. ISBN 978-0-6482278-5-4. <https://www.eoa.org.au/earth-observation-textbooks>. Australia and New Zealand CRC for Spatial Information.
- Bennett, S., C. M. Duarte, N. Marbà, and T. Wernberg. 2019. Integrating within-species variation in thermal physiology into climate change ecology. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **374**: 20180550. doi:10.1098/rstb.2018.0550
- Bennett, S., and others. 2022. Resilience of seagrass populations to thermal stress does not reflect regional differences in ocean climate. *New Phytol.* **233**: 1657–1666. doi:10.1111/nph.17885
- Calvo, S., G. Lovison, M. Pirrotta, G. Di Maida, A. Tomasello, and M. Sciandra. 2006. Modelling the relationship between sexual reproduction and rhizome growth in *Posidonia oceanica* (L.) Delile. *Mar. Ecol.* **27**: 361–371. doi:10.1111/j.1439-0485.2006.00132.x
- Campagne, C. S., J.-M. Salles, P. Boissery, and J. Deter. 2015. The seagrass *Posidonia oceanica*: Ecosystem services identification and economic evaluation of goods and benefits. *Mar. Pollut. Bull.* **97**: 391–400. doi:10.1016/j.marpolbul.2015.05.061
- Chefaoui, R. M., C. M. Duarte, and E. A. Serrão. 2017. Palaeoclimatic conditions in the Mediterranean explain genetic diversity of *Posidonia oceanica* seagrass meadows. *Sci. Rep.* **7**: 2732. doi:10.1038/s41598-017-03006-2
- Coll, M., and others. 2010. The biodiversity of the Mediterranean Sea: Estimates, patterns, and threats. *PLoS One* **5**: e11842. doi:10.1371/journal.pone.0011842
- Coma, R., M. Ribes, E. Serrano, E. Jiménez, J. Salat, and J. Pascual. 2009. Global warming-enhanced stratification and mass mortality events in the Mediterranean. *Proc. Natl. Acad. Sci.* **106**: 6176–6181. doi:10.1073/PNAS.0805801106
- Darmaraki, S., and others. 2019. Future evolution of marine heatwaves in the Mediterranean Sea. *Climate Dynam.* **53**: 1371–1392. doi:10.1007/s00382-019-04661-z
- Díaz-Almela, E., N. Marbà, E. Álvarez, E. Balestri, J. M. Ruiz-Fernández, and C. M. Duarte. 2006. Patterns of seagrass (*Posidonia oceanica*) flowering in the Western Mediterranean. *Mar. Biol.* **148**: 723–742. doi:10.1007/s00227-005-0127-x
- Díaz-Almela, E., N. Marbà, and C. M. Duarte. 2007. Consequences of Mediterranean warming events in seagrass (*Posidonia oceanica*) flowering records. *Glob. Chang. Biol.* **13**: 224–235. doi:10.1111/j.1365-2486.2006.01260.x
- Díaz-Almela, E., N. Marbà, R. Martínez, R. Santiago, and C. M. Duarte. 2009. Seasonal dynamics of *Posidonia oceanica* in Magalluf Bay (Mallorca, Spain): Temperature effects on seagrass mortality. *Limnol. Oceanogr.* **54**: 2170–2182. doi:10.4319/llo.2009.54.6.2170
- Entrambasaguas, L., M. Jahnke, E. Biffali, M. Borra, R. Sanges, L. Marín-Guirao, and G. Procaccini. 2017. Tissue-specific transcriptomic profiling provides new insights into the reproductive ecology and biology of the iconic seagrass species *Posidonia oceanica*. *Mar. Genom.* **35**: 51–61. doi:10.1016/j.margen.2017.05.006
- Frölicher, T. L., E. M. Fischer, and N. Gruber. 2018. Marine heatwaves under global warming. *Nature* **560**: 360–364. doi:10.1038/s41586-018-0383-9
- Garrabou, J., and others. 2009. Mass mortality in northwestern Mediterranean rocky benthic communities: Effects of the 2003 heat wave. *Glob. Chang. Biol.* **15**: 1090–1103. doi:10.1111/j.1365-2486.2008.01823.x
- Garrabou, J., and others. 2022. Marine heatwaves drive recurrent mass mortalities in the Mediterranean Sea. *Glob. Chang. Biol.* **28**: 5708–5725. doi:10.1111/gcb.16301
- Gerakaris, V., V. Papathanasiou, M. Salomidi, Y. Issaris, and P. Panayotidis. 2021. Spatial patterns of *Posidonia oceanica* structural and functional features in the eastern Mediterranean (Aegean and E. Ionian seas) in relation to large-scale environmental factors. *Mar. Environ. Res.* **165**: 105222. doi:10.1016/j.marenvres.2020.105222
- Hemminga, M. A., and C. M. Duarte. 2000. *Seagrass ecology*. Cambridge University Press.
- Hobday, A. J., and others. 2016. A hierarchical approach to defining marine heatwaves. *Prog. Oceanogr.* **141**: 227–238. doi:10.1016/j.pocean.2015.12.014
- Hobday, A., and others. 2018. Categorizing and naming marine heatwaves. *Oceanography* **31**: 162–173. doi:10.5670/oceanog.2018.205
- Hughes, R. A., and J. J. Stachowicz. 2004. Genetic diversity enhances the resistance of a seagrass ecosystem to disturbance. *PNAS* **101**: 8998–9002. doi:10.1073/pnas.0402642101
- Jahnke, M., J. F. Pagès, T. Alcoverro, P. S. Lavery, K. M. McMahon, and G. Procaccini. 2015. Should we sync? Seascape-level genetic and ecological factors determine seagrass flowering patterns N. Rafferty [ed.]. *J. Ecol.* **103**: 1464–1474. doi:10.1111/1365-2745.12470
- Juza, M., and J. Tintoré. 2021. Multivariate sub-regional ocean indicators in the Mediterranean Sea: From event detection

- to climate change estimations. *Front. Mar. Sci.* **8**: 233. doi:[10.3389/FMARS.2021.610589/BIBTEX](https://doi.org/10.3389/FMARS.2021.610589/BIBTEX)
- Kendrick, G. A., and others. 2012. The central role of dispersal in the maintenance and persistence of seagrass populations. *Bio-science* **62**: 56–65. doi:[10.1525/bio.2012.62.1.10](https://doi.org/10.1525/bio.2012.62.1.10)
- Knies, J. L., J. G. Kingsolver, and C. L. Burch. 2009. Hotter is better and broader: Thermal sensitivity of fitness in a population of bacteriophages. *Am. Nat.* **173**: 419–430. doi:[10.1086/597224](https://doi.org/10.1086/597224)
- Leaver, M., E. Moreno, M. Kayhan, A. McGaughran, C. Rödelserperger, R. J. Sommer, and A. A. Hyman. 2022. Adaptation to environmental temperature in divergent clades of the nematode *Pristionchus pacificus*. *Evolution* **76**: 1660–1673. doi:[10.1111/evo.14520](https://doi.org/10.1111/evo.14520)
- Litsi-Mizan, V., P. T. Efthymiadis, V. Gerakaris, O. Serrano, M. Tsapakis, and E. T. Apostolaki. 2023. Decline of seagrass (*Posidonia oceanica*) production over two decades in the face of warming of the eastern Mediterranean Sea. *New Phytol.* **239**: 2126–2137. doi:[10.1111/nph.19084](https://doi.org/10.1111/nph.19084)
- Litsi-Mizan, V., C. A. García-Escudero, C. S. Tsigenopoulos, K. Tsiaras, V. Gerakaris, and E. T. Apostolaki. 2024. Unravelling the genetic pattern of seagrass (*Posidonia oceanica*) meadows in the eastern Mediterranean Sea. *Biodivers. Conserv.* **33**: 257–280. doi:[10.1007/s10531-023-02746-0](https://doi.org/10.1007/s10531-023-02746-0)
- Marbà, N., R. Santiago, E. Díaz-Almela, E. Álvarez, and C. M. Duarte. 2006. Seagrass (*Posidonia oceanica*) vertical growth as an early indicator of fish farm-derived stress. *Estuar. Coast. Shelf Sci.* **67**: 475–483. doi:[10.1016/j.ecss.2005.11.034](https://doi.org/10.1016/j.ecss.2005.11.034)
- Marbà, N., and C. M. Duarte. 2010. Mediterranean warming triggers seagrass (*Posidonia oceanica*) shoot mortality. *Glob. Chang. Biol.* **16**: 2366–2375. doi:[10.1111/j.1365-2486.2009.02130.x](https://doi.org/10.1111/j.1365-2486.2009.02130.x)
- Marbà, N., G. Jordà, S. Agustí, C. Girard, and C. M. Duarte. 2015. Footprints of climate change on Mediterranean Sea biota. *Front. Mar. Sci.* **2**: 1–11. doi:[10.3389/fmars.2015.00056](https://doi.org/10.3389/fmars.2015.00056)
- Marín-Guirao, L., J. M. Ruiz, E. Dattolo, R. Garcia-Munoz, and G. Procaccini. 2016. Physiological and molecular evidence of differential short-term heat tolerance in Mediterranean seagrasses. *Sci. Rep.* **6**: 28615. doi:[10.1038/srep28615](https://doi.org/10.1038/srep28615)
- Marín-Guirao, L., L. Entrambasaguas, J. M. Ruiz, and G. Procaccini. 2019. Heat-stress induced flowering can be a potential adaptive response to ocean warming for the iconic seagrass *Posidonia oceanica*. *Mol. Ecol.* **28**: 2486–2501. doi:[10.1111/mec.15089](https://doi.org/10.1111/mec.15089)
- Martinez, J., F. Leonelli, E. García-Ladona, J. Garrabou, D. Kersting, N. Bensoussan, and A. Pisano. 2023. Evolution of marine heatwaves in warming seas: The Mediterranean Sea case study. *Front. Mar. Sci.* **10**: 1193164. doi:[10.3389/fmars.2023.1193164](https://doi.org/10.3389/fmars.2023.1193164)
- Mayot, N., C.-F. Boudouresque, and A. Leriche. 2005. Unexpected response of the seagrass *Posidonia oceanica* to a warm-water episode in the North Western Mediterranean Sea. *C. R. Biol.* **328**: 291–296. doi:[10.1016/j.crvi.2005.01.005](https://doi.org/10.1016/j.crvi.2005.01.005)
- McDonald, A. M., R. B. McDonald, J. Cebrian, and J. L. Sánchez Lizaso. 2023. Reconstructed life history metrics of the iconic seagrass *Posidonia oceanica* (L.) detect localized anthropogenic disturbance signatures. *Mar. Environ. Res.* **186**: 105901. doi:[10.1016/j.marenvres.2023.105901](https://doi.org/10.1016/j.marenvres.2023.105901)
- Montefalcone, M., E. Giovannetti, C. Morri, A. Peirano, and C. N. Bianchi. 2013. Flowering of the seagrass *Posidonia oceanica* in the NW Mediterranean: Is there a link with solar activity? *Medit. Mar. Sci.* **14**: 416. doi:[10.12681/mms.529](https://doi.org/10.12681/mms.529)
- Nguyen, H. M., P. J. Ralph, L. Marín-Guirao, M. Pernice, and G. Procaccini. 2021. Seagrasses in an era of ocean warming: A review. *Biol. Rev.* **96**: 2009–2030. doi:[10.1111/BRV.12736](https://doi.org/10.1111/BRV.12736)
- Nguyen, H. M., and others. 2023. Signs of local adaptation by genetic selection and isolation promoted by extreme temperature and salinity in the Mediterranean seagrass *Posidonia oceanica*. *Mol. Ecol.* **32**: 4313–4328. doi:[10.1111/mec.17032](https://doi.org/10.1111/mec.17032)
- Olesen, B., N. Marba, C. M. Duarte, R. S. Savela, and M. D. Fortes. 2004. Recolonization dynamics in a mixed seagrass meadow: The role of clonal versus sexual processes. *Estuaries* **27**: 770–780. doi:[10.1007/BF02912039](https://doi.org/10.1007/BF02912039)
- Oliver, E. C. J., and others. 2018. Longer and more frequent marine heatwaves over the past century. *Nat. Commun.* **9**: 1324. doi:[10.1038/s41467-018-03732-9](https://doi.org/10.1038/s41467-018-03732-9)
- Oliver, E. C. J., and others. 2019. Projected marine heatwaves in the 21st century and the potential for ecological impact. *Front. Mar. Sci.* **6**: 734. doi:[10.3389/fmars.2019.00734](https://doi.org/10.3389/fmars.2019.00734)
- Panayotidis, P., and others. 2022. Seagrass meadows in the Greek seas: Presence, abundance and spatial distribution. *Bot. Mar.* **65**: 289–299. doi:[10.1515/bot-2022-0011](https://doi.org/10.1515/bot-2022-0011)
- Pastor, F., J. A. Valiente, and S. Khodayar. 2020. A warming Mediterranean: 38 years of increasing sea surface temperature. *Remote Sens. (Basel)* **12**: 2687. doi:[10.3390/rs12172687](https://doi.org/10.3390/rs12172687)
- Pastor, F., and S. Khodayar. 2023. Marine heat waves: Characterizing a major climate impact in the Mediterranean. *Sci. Total Environ.* **861**: 160621. doi:[10.1016/j.scitotenv.2022.160621](https://doi.org/10.1016/j.scitotenv.2022.160621)
- Peirano, A. 2002. Lepidochronology and internodal length methods for studying *Posidonia oceanica* growth: Are they compatible? *Aquat. Bot.* **74**: 175–180. doi:[10.1016/S0304-3770\(02\)00078-5](https://doi.org/10.1016/S0304-3770(02)00078-5)
- Pergent, G., C.-F. Boudouresque, A. Crouzet, and A. Meinesz. 1989. Cyclic changes along *Posidonia oceanica* rhizomes (Lepidochronology): Present state and perspectives. *Mar. Ecol.* **10**: 221–230. doi:[10.1111/j.1439-0485.1989.tb00474.x](https://doi.org/10.1111/j.1439-0485.1989.tb00474.x)
- Pergent, G. 1990. Lepidochronological analysis of the seagrass *Posidonia oceanica* (L.) Delile: A standardized approach. *Aquat. Bot.* **37**: 39–54. doi:[10.1016/0304-3770\(90\)90063-Q](https://doi.org/10.1016/0304-3770(90)90063-Q)
- Pisano, A., S. Marullo, V. Artale, F. Falcini, C. Yang, F. E. Leonelli, R. Santoleri, and B. B. Nardelli. 2020. New evidence of Mediterranean climate change and variability from sea surface temperature observations. *Remote Sens. (Basel)* **12**: 1–18. doi:[10.3390/RS12010132](https://doi.org/10.3390/RS12010132)
- Planes, S., N. Raventos, B. Ferrari, and T. Alcoverro. 2011. Fish herbivory leads to shifts in seagrass *Posidonia oceanica* investments in sexual reproduction. *Mar. Ecol. Prog. Ser.* **431**: 205–213. doi:[10.3354/meps09089](https://doi.org/10.3354/meps09089)

- Procaccini, G., J. L. Olsen, and T. B. H. Reusch. 2007. Contribution of genetics and genomics to seagrass biology and conservation. *J. Exp. Mar. Biol. Ecol.* **350**: 234–259. doi:10.1016/j.jembe.2007.05.035
- R Core Team. 2022. R: A language and environment for statistical computing. R Foundation for Statistical Computing, p. 2012.
- Ruiz, J. M., and J. Romero. 2003. Effects of disturbances caused by coastal constructions on spatial structure, growth dynamics and photosynthesis of the seagrass *Posidonia oceanica*. *Mar. Pollut. Bull.* **46**: 1523–1533. doi:10.1016/j.marpolbul.2003.08.021
- Ruiz, J. M., and others. 2018. Experimental evidence of warming-induced flowering in the Mediterranean seagrass *Posidonia oceanica*. *Mar. Pollut. Bull.* **134**: 49–54. doi:10.1016/j.marpolbul.2017.10.037
- Schlegel, R. W., and A. J. Smit. 2018. heatwaveR: A central algorithm for the detection of heatwaves and cold-spells. *JOSS* **3**: 821. doi:10.21105/joss.00821
- Serra, I. A., and others. 2010. Genetic structure in the Mediterranean seagrass *Posidonia oceanica*: Disentangling past vicariance events from contemporary patterns of gene flow. *Mol. Ecol.* **19**: 557–568. doi:10.1111/j.1365-294X.2009.04462.x
- Short, F., T. Carruthers, W. Dennison, and M. Waycott. 2007. Global seagrass distribution and diversity: A bioregional model. *J. Exp. Mar. Biol. Ecol.* **350**: 3–20. doi:10.1016/j.jembe.2007.06.012
- Smale, D. A., and others. 2019. Marine heatwaves threaten global biodiversity and the provision of ecosystem services. *Nat. Clim. Change* **9**: 306–312. doi:10.1038/s41558-019-0412-1
- Smith, K., and others. 2023. Biological impacts of marine heatwaves. *Ann. Rev. Mar. Sci.* **15**: 1–27. doi:10.1146/annurev-marine-032122-121437
- Stipich, P., and others. 2022. Assessment of *Posidonia oceanica* traits along a temperature gradient in the Mediterranean Sea shows impacts of marine warming and heat waves. *Front. Mar. Sci.* **9**: 895354. doi:10.3389/fmars.2022.895354
- Stipich, P., and others. 2023. Effects of high temperature and marine heat waves on seagrasses: Is warming affecting the nutritional value of *Posidonia oceanica*? *Mar. Environ. Res.* **184**: 105854. doi:10.1016/j.marenvres.2022.105854
- Stipich, P., G. La Manna, and G. Ceccherelli. 2024. Warming-induced flowering and fruiting in the seagrass *Posidonia oceanica* and uncertainties due to context-dependent features. *Mar. Biol.* **171**: 67. doi:10.1007/s00227-024-04388-4
- Tomas, F., Hernan, G., Mañez-Crespo, J., Arona, A., Meléndez, D. H., Reynés, X., Delgado, J., Procaccini, G., & Ballesteros, E. 2024. Mass flowering and unprecedented extended pseudovivipary in seagrass (*Posidonia oceanica*) after a Marine Heat Wave. *Mar. Pollut. Bull.* **203**: 116394. doi:10.1016/j.marpolbul.2024.116394
- Tomasello, A., M. Sciandra, V. M. R. Muggeo, M. Pirrotta, G. Di Maida, and S. Calvo. 2016. Reference growth charts for *Posidonia oceanica* seagrass: An effective tool for assessing growth performance by age and depth. *Ecol. Indic.* **69**: 50–58. doi:10.1016/j.ecolind.2016.04.005
- Tutar, O., M. Ruocco, E. Dattolo, and others. 2022. High levels of genetic diversity and population structure in the Mediterranean seagrass *Posidonia oceanica* at its easternmost distribution limit. *CES J. Mar. Sci.* **79**: 2286–2297. doi:10.1093/icesjms/fsac163
- Van Drunen, W. E., and M. E. Dorken. 2012. Trade-offs between clonal and sexual reproduction in *Sagittaria latifolia* (Alismataceae) scale up to affect the fitness of entire clones. *New Phytol.* **196**: 606–616. doi:10.1111/j.1469-8137.2012.04260.x
- Waycott, M., G. Procaccini, D. H. Les, and T. B. H. Reusch. 2006. Seagrass evolution, ecology and conservation: A genetic perspective, p. 25–50. In A. W. D. Larkum, R. J. orth, and C. M. Duarte [eds.], *Seagrasses: Biology, ecology and conservation*. Springer.
- Wickham, H. 2016. ggplot2: Elegant graphics for data analysis.
- Wood, S. N. 2011. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **73**: 3–36. doi:10.1111/j.1467-9868.2010.00749.x
- Wood, S. N., F. Scheipl, and M. S. Wood. 2017. Package ‘gamm4’. *Am Stat* **45**: 10.

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Conflict of Interest

All co-authors have seen and agreed with the contents of the manuscript and there are no relevant financial or non-financial interests to disclose.

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